

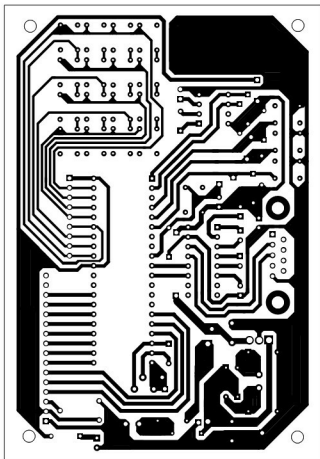


Fig. 2: An actual-size, single-side PCB for the circuit of RTC-based event logger

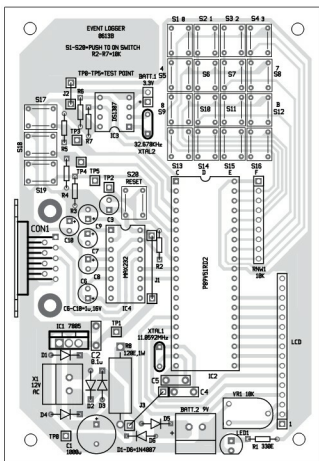


Fig. 3: Component layout for the PCB

the last 100 entries will appear on the HyperTerminal.

Construction and testing

An actual-size, single-side PCB for the circuit of the event logger (Fig. 1) is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it saves time and minimises assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC bases for the microcontroller and the RTC. Before inserting the ICs, check the supply voltage. Using preset VR1 set the LCD contrast for proper display. To test the circuit for proper working, check the voltages at different points as per Table II.

Software

The source program for the micro-controllers is written in 'C' language and compiled using Keil μ Vision4

compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. The program is well-commented and easy to understand.

On reset/power-'on,' the micro-controller executes the main program. It first initialises variables, baud rate and the LCD, then starts displaying the messages for setting the time and date using start() function. Time and date are to be entered from the numeric keypad. Key_scan() function scans for key entries. Message 'Invalid Key' is displayed on the LCD if any wrong key is pressed. A continuous loop always displays the date and time and checks the status of switches S17 through S19. Function DS1307_get(unsigned char addr) is used to read the time and date, which are displayed on the LCD by function sendlcd(unsigned char value).

When an event occurs (switch S17

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

pressed), the time and date of the event are saved on the microcontroller's memory location by data_shift() function and message "You are logged" is displayed on the LCD. Switches S18 and S19 are used to display the last ten recorded data on the LCD one by one and the last 100 recorded data on the HyperTerminal. Functions DS1307_settime(unsigned char hh, unsigned char min, unsigned char ss) and DS1307_setdate(unsigned char dd, unsigned char mm, unsigned char yy) are used to set the time and date, respectively. ●

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